UNIT 1 · TOPIC 1.5

Leveraging AI in Cyber Defense

Quiz

Instructions: This quiz has 18 multiple-choice questions, 1 point each. Circle the letter of the single best answer for each question.

Name: **Date:** **Class:**

1. Which of the following BEST describes the primary function of an AI-powered Intrusion Prevention System (IPS)?

- ○ A. A firewall that blocks traffic based solely on a static list of known bad IP addresses
- ○ B. A security control that uses machine learning to detect and block malicious network traffic in real time
- ○ C. A tool that encrypts data at rest to protect its confidentiality from unauthorized access
- ○ D. A system that logs all user login attempts without taking any automated action

2. In the scenario, the developer uses an AI tool to review application code and flag security vulnerabilities before the web application is launched. This is an example of which type of security control?

- ○ A. Preventive, it identifies and addresses vulnerabilities before they can be exploited
- ○ B. Corrective, it responds to and remedies an incident after it has occurred
- ○ C. Detective, it identifies attacks as they happen during normal operation
- ○ D. Compensating, it substitutes for another control that cannot be implemented

3. Which of the following is MOST accurately described as an AI-assisted security control for protecting application data from injection attacks?

- ○ A. A password policy requiring all users to create passwords of at least eight characters
- ○ B. A physical lock on the server room door to prevent unauthorized hardware access
- ○ C. An AI tool that validates and sanitizes user input before passing it to a database query
- ○ D. A scheduled weekly backup of the database to an off-site location

4. A software team is building an e-commerce site that connects to a product inventory database. An AI code-review tool flags five input fields where user-supplied data is passed directly into SQL queries without any validation. How does implementing the tool's recommended input validation MOST directly mitigate the identified risk?

- ○ A. It encrypts the database contents so adversaries cannot read the data even if they gain access
- ○ B. It limits the number of failed login attempts before automatically locking out a user account
- ○ C. It monitors all inbound network traffic to the server and blocks packets matching known attack signatures
- ○ D. It prevents adversaries from submitting crafted input that could manipulate the structure of database queries

5. A company uses an AI-powered Security Information and Event Management (SIEM) system that correlates logs from ten systems. It flags a pattern where a privileged account is accessing payroll files at 2:00 AM every night, though the account owner has never worked night shifts. Which security goal does the SIEM alert MOST directly support?

- ○ A. Ensuring the availability of payroll files for authorized users during business hours
- ○ B. Encrypting payroll data to protect its confidentiality from unauthorized disclosure
- ○ C. Identifying potential insider threat activity before data is exfiltrated or damaged

○ D. Preventing all after-hours access to the payroll system through a blanket access block

6. An organization wants to reduce the time between when a new vulnerability is discovered in its web server software and when a patch is deployed. Which AI-assisted security control BEST addresses this specific goal?

○ A. A physical security camera installed near the server rack to deter unauthorized hardware access

○ B. A managerial policy requiring IT staff to review vendor patch notes on a quarterly schedule

○ C. An AI-powered vulnerability scanner that continuously monitors public CVE databases and automatically alerts the team when relevant new vulnerabilities are published

○ D. A password manager enforcing multi-factor authentication for all administrative accounts

7. A cybersecurity team debates whether to replace their entire security stack with a single AI-powered platform promising unified threat prevention, detection, and response. A senior analyst argues against it, citing defense-in-depth principles. Which reasoning BEST supports the analyst's position?

○ A. AI security platforms are generally too expensive for most organizations to deploy as a complete solution

○ B. Current AI tools cannot aggregate and correlate logs from more than one data source at a time

○ C. Human security analysts are always more accurate than AI-powered tools at detecting advanced persistent threats

○ D. A single unified platform creates a monoculture, if the AI model is evaded or fails, all three security functions fail simultaneously

8. In the scenario, an AI code-review tool recommends code changes to address injection vulnerabilities. The development team implements all recommended changes without review and pushes them directly to production. Which risk does this approach MOST significantly introduce?

○ A. The AI tool may have identified fewer vulnerabilities than a full manual penetration test would have found

○ B. Customers will be unable to place orders until the AI tool re-reviews the updated code before deployment

○ C. Input validation routines will significantly degrade database query performance under high traffic loads

○ D. Unreviewed AI-generated code changes could introduce new logical errors or unintended behaviors into the production system

9. An organization deploys three layered AI-powered controls: (1) an AI IDS monitoring inbound network traffic, (2) an AI code-review tool for all application deployments, and (3) an AI SIEM correlating logs across all systems. A new ransomware strain evades the IDS by using encrypted command-and-control channels, but the SIEM detects anomalous outbound traffic volume from an infected endpoint. What does this outcome BEST demonstrate about the organization's AI security architecture?

○ A. The AI IDS is misconfigured and should be decommissioned and replaced with a human security analyst

○ B. The SIEM is the only AI tool worth deploying since it detected the threat that the IDS missed

○ C. Layered AI controls can compensate for individual tool failures when each tool operates on different evidence sources and detection methods

○ D. Encrypted command-and-control traffic is undetectable by any security control and should be accepted as an unavoidable residual risk

10. Which of the following BEST describes how an AI-powered Intrusion Detection System (IDS) monitors network traffic to identify threats?

○ A. It scans the database server for unauthorized changes to stored records at scheduled intervals

○ B. It encrypts all outbound network packets to prevent an adversary from reading transmitted data

○ C. It automatically locks out user accounts after a pre-configured number of failed authentication attempts

○ D. It learns a baseline of normal network behavior through training and flags traffic that deviates significantly from that baseline

11. In the scenario, the AI code-review tool produces a report flagging specific input fields as potential injection vulnerabilities. What type of evidence does this report represent in the context of a pre-deployment security review?

○ A. Application-layer findings identifying specific locations in source code where user input is handled unsafely

○ B. A physical access log recording which personnel entered the server room during development

○ C. A network packet capture recording malicious traffic targeting the application in real time

○ D. A browser history file showing which websites the development team accessed during the project

12. Which monitoring method would MOST likely detect a previously unknown (zero-day) malware variant that has no known signature in any existing threat database?

○ A. Signature-based antivirus scanning comparing files against a database of known malware hashes

○ B. AI-powered anomaly detection monitoring for unusual system behaviors such as unexpected process creation or abnormal outbound connections

○ C. Static perimeter blacklist filtering blocking traffic from a list of known malicious IP addresses

○ D. Scheduled weekly scans using the most recently updated malware definition database

13. An AI SIEM analyzes login logs and detects the following for a single administrator account: successful logins from the same US-based IP address between 9 AM and 5 PM on weekdays for six months, followed by a successful login from a foreign IP address at 3 AM local time. The SIEM flags this as anomalous. Which principle BEST explains why this access pattern is suspicious?

○ A. The foreign login used an incorrect password multiple times, which is consistent with a brute-force credential attack

○ B. Logins originating from foreign IP addresses are universally blocked by standard network security policy

○ C. The significant deviation from an established behavioral baseline suggests the account credentials may have been compromised

○ D. Administrator accounts should never be accessed during off-hours under any operational circumstances

14. After the AI code-review tool flags injection vulnerabilities in the scenario's web application, the development team wants to verify that their fixes actually work before deploying to production. Which monitoring approach would generate the MOST useful evidence of whether the remediation was successful?

○ A. Run automated penetration tests in the test environment that simulate injection attacks and observe whether they succeed or fail

○ B. Review the server's physical access logs to confirm only authorized developers made changes to the codebase

○ C. Review the company's password policy to confirm all developer accounts require strong credentials

○ D. Monitor the office building's CCTV footage during the testing period to confirm no unauthorized personnel were present

15. An AI monitoring system flags an endpoint that is sending small, regular outbound packets to an unrecognized external IP address every 15 minutes, a pattern consistent with command-and-control (C2) beaconing. The endpoint's user reports noticing nothing unusual. Which action BEST represents sound evidence collection for the incident response team?

○ A. Reset the endpoint user's account password and close the incident ticket as resolved

○ B. Capture full network packets from the endpoint and cross-reference the external IP address against threat intelligence feeds

○ C. Ask the user to restart their machine and monitor whether the outbound packets stop after the reboot

○ D. Disable the AI monitoring system until a human analyst can manually review the alert queue

16. An AI monitoring tool generates 2,400 security alerts in a single 24-hour period. A security analyst samples the alerts and finds that 85% are false positives, legitimate activity incorrectly flagged, while the remaining 15% include three genuine high-severity incidents buried among hundreds of low-severity true positives. Which operational change MOST directly and appropriately addresses the root cause of this problem?

○ A. Disable the AI monitoring tool entirely and return to manual log review by the security team

○ B. Retrain or tune the AI model using labeled examples of true positives and false positives to improve detection precision

○ C. Raise the alert severity threshold so the system only generates alerts for events it classifies as critical severity

○ D. Hire additional analysts to manually triage all 2,400 daily alerts generated by the existing system

17. An incident response team receives the following four AI-generated security alerts simultaneously. Evaluate which alert should be prioritized FIRST for investigation and response.

○ A. An AI phishing filter quarantined 12 incoming emails from a known bulk-mail domain before any user opened them

○ B. An AI vulnerability scanner identified a medium-severity CVE in a third-party library used by the internal development server

○ C. An AI IDS detected an internal production server transmitting 40 GB of data to an unrecognized external IP address between 2:00 AM and 3:00 AM

○ D. An AI SIEM flagged a single failed login attempt on an employee's account originating from the employee's registered home IP address

18. The scenario describes a human development team reviewing AI-generated code recommendations before pushing changes to production. Evaluate which statement BEST identifies the primary security benefit of this human-in-the-loop step AND correctly identifies the condition that would most directly undermine it.

○ A. Benefit: Human developers review code faster than AI tools can generate recommendations; undermined by: developers and the AI tool using incompatible integrated development environments

○ B. Benefit: Human developers can evaluate business logic, deployment context, and edge cases the AI tool may not understand; undermined by: automatically accepting all AI recommendations without independent code review

○ C. Benefit: Human review guarantees that no security vulnerability will reach the production environment; undermined by: the organization deploying more than one AI code-review tool simultaneously

○ D. Benefit: Human review allows the AI tool to incorporate corrections into its model and improve its future recommendations; undermined by: development team members working across multiple time zones